Analysis Method for Downlink Co-Frequency Interference in NGSO Satellite Communication Systems With Information Geometry

Yuanzhi He, Di Yan, and Chengwu Qi

***Abstract*—The rapid expansion of mega Non-Geostationary Orbit (NGSO) constellation systems has aggravated downlink co-frequency interference (CFI) problem. This work proposes an interference analysis method based on the "distance to center" (DTC) of manifold to analyze CFI in NGSO constellation systems. We establish the manifold geometric structure and parametric characterization model for the probability distribution of the downlink CFI between NGSO systems. The performance of the proposed method under two metrics, Affine Invariant Riemannian Metric (AIRM) and Symmetric Kullback-Leibler Divergence (SKLD), in scenarios involving potential interference from two NGSO satellites is analyzed through simulation, and compared with traditional energy-based method. The results show that the proposed method can effectively mitigate the complexity of interference analysis caused by the time-varying characteristics of NGSO constellations. Additionally, by mining high-dimensional information from signals, interference can be rapidly and accurately analyzed, which provides a novel and efficient method for the analysis of the CFI between NGSO constellation systems and system link design. Finally, the interference signals received by NGSO Earth station are visualized in 3D space by affine embedding method, which can provide an intuitive reference for interference analysis between NGSO systems.**

***Index Terms*—NGSO constellation, co-frequency interference, information geometry, interference analysis.**

I. Introduction

IN RECENT years, the low Earth orbit (LEO) satellite communication systems based on the mega NGSO constellation have experienced rapid development, and communication operators and emerging technology companies in various countries have put forward their own mega NGSO constellation plans and carried out vigorous construction. Among them, representative ones include SpaceX's "Starlink" constellation and Amazon's "Kuiper" constellation of the United States, the "OneWeb" constellation of the United Kingdom, the "Telesat" constellation of Canada and the "GW" constellation of China [1], [2], [3], [4]. Due to the limited space frequency resources, the communication frequency bands that the International Telecommunication Union (ITU) can allocate to these emerging NGSO constellations will inevitably overlap, which leads to the increasing problem of co-frequency interference (CFI) between them [5], [6], [7], [8]. The detection and analysis of the CFI between NGSO constellations have become a focal points in the field of satellite communication.

However, it is found in the research that there are many difficulties in the detection and analysis of the CFI between NGSO constellations:

- The NGSO constellation(especially the mega NGSO constellation) has a large number of satellites, which leads to the results that the number of interference sources that should be considered is large and the interference scenarios are very complex;
- The NGSO satellite travels around the Earth at a very high speed, resulting in the constant change of the relative positions of satellites in different orbital planes. Consequently, the interference amplitude and interference direction change rapidly over time;
- The communication frequency bands of NGSO constellations are evolving to higher frequencies such as Ku and Ka, and the analysis of CFI spans a wider frequency range, thereby increasing the complexity of the analysis.

Therefore, how to achieve rapid and precise detection and analysis of CFI among NGSO constellation systems is a difficult problem that needs to be solved urgently [9], [10], [11], [12].

Traditional research often constructs static interference models based on geometric characteristics or link parameters. For example, the authors of [13] revealed the influence of interference angles between NGSO and GEO systems through bit error rate performance analysis. However, the model focuses solely on static angular factors and does not incorporate time-varying orbital dynamic parameters, limiting the comprehensiveness of interference assessment. The authors of [14] proposed the exclusive angle strategy, which reduces downlink interference to GEO systems by restricting the coverage of NGSO satellites. However, this approach sacrifices NGSO service capabilities and does not quantify the frequency-domain and power-domain characteristics of interference signals. The authors of [15] drew on GEO system protection standards to propose a harmful interference probability model for NGSO constellations and verifies the effectiveness of Ka-band uplink interference suppression through a spatial isolation

Received 2 December 2024; revised 23 March 2025; accepted 23 April 2025. Date of publication 8 May 2025; date of current version 14 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62341128. The associate editor coordinating the review of this article and approving it for publication was G. Chen. *(Corresponding author: Yuanzhi He.)*

Yuanzhi He is with PLA, Beijing 100141, China (e-mail: he_yuanzhi@126.com).

Di Yan and Chengwu Qi are with the Institute of Systems Engineering, Academy of Military Sciences, Beijing 100141, China (e-mail: yandi_2022@126.com; chengwu_qi@163.com).

Digital Object Identifier 10.1109/TWC.2025.3566234

1536-1276 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

angle coordination mechanism. However, its applicability is limited to specific frequency band scenarios. With the expansion of mega-constellation deployments, the spatiotemporal dynamics of interference have become a research focus. The authors of [16] and [17] made progress in this direction: the former proposes a link angle probability analysis method and constellation availability metrics but does not delve into the time-varying characteristics of interference; the latter innovatively introduces spatiotemporal slicing technology, decomposing interference scenarios into time segments and spatial grids, and reveals the spatiotemporal distribution of CFI from NGSO system to GEO system through global characteristic analysis. Similarly, the authors of [18] designed a system-level capacity loss metric to simulate the interference probability distribution of hybrid NGSO constellations, but the model does not analyze the interference coupling mechanisms at the signal level. For multi-orbit system coexistence scenarios, the authors of [19] and [20] proposed solutions from the perspectives of spatial isolation and beam optimization, respectively. The authors of [19] verified the correlation between isolation angles, GEO protection zone width, and NGSO constellation density through simulations, but its conclusions rely on preset static density parameters, making it difficult to adapt to dynamic constellation expansion needs. The authors of [20] established a multi-beam interference analysis model and proposes a co-frequency cell determination method based on coordinate transformation, effectively evaluating the impact of constellation scale on uplink interference but not addressing cross-system (e.g., NGSO-GEO) interference coordination. Meanwhile, the authors of [21] established a unified multi-beam analysis model for interference between NGSO satellite communication systems to conduct downlink interference analysis, but its computational efficiency is limited by the scale of full-scenario simulations.

The aforementioned traditional interference analysis methods primarily analyze interference from perspectives such as link angles, extremum of interference functions, slicing, or partitioning. At their core, these methods focus on detecting and analyzing the energy of signals, lacking an intuitive representation of interference signals. They do not consider the geometric characteristics of the signals themselves, making it difficult to effectively extract high-dimensional information from the received signals. To address these limitations, this paper proposes a CFI analysis method for NGSO satellite communication systems based on information geometry. By fully considering the time-varying characteristics of interference between NGSO constellations, the method models the CFI situation between NGSO constellations on a statistical manifold using the probability distribution characteristics of the sampled signals at the receiver. This approach endows the interference signals and their underlying high-dimensional information with a geometric structure, enabling an intuitive representation of interference signals. Additionally, real-time and accurate interference detection and analysis are achieved through geometric computations on the statistical manifold.

Information geometry is a rapidly emerging interdisciplinary field that combines information theory, probability theory, and differential geometry. It studies problems in statistics and information theory using modern differential geometric methods based on Riemannian manifolds. By considering a family of probability distributions as a whole, it deeply explores the geometric structure information contained within the family of probability distribution functions. Information geometry can transform many concepts in statistics and information theory into geometric structures in manifold space, and then study their properties using differential geometric methods, thus turning complex algebraic problems into geometric problems on manifolds. This provides a new method and perspective for solving problems such as signal detection, parameter estimation, and interference analysis [22], [23], [24], [25]. This paper fully utilizes the advantages of information geometry in characterizing the statistical properties of time-varying signals, mining high-dimensional information, and performing calculations on statistical manifolds. It applies information geometry to solve the problem of interference analysis and detection in mega NGSO constellations, thereby overcoming the shortcomings of traditional interference analysis methods, which are insufficient in considering signal geometric characteristics, mining high-dimensional information, representing interference signals, and making real-time judgments.

The main contributions of this paper are summarized as follows:

- A comprehensive transmission model is established, along with the manifold geometric structure and parametric representation of the probability distribution of signals received by Earth stations in the scenario of downlink CFI between NGSO constellations.
- A manifold "distance to center" (DTC) based interference analysis method is proposed. Its performance is rigorously evaluated under the AIRM and SKLD metric, utilizing orbital data from the "OneWeb" and "Starlink" constellations.
- The detection probability of the proposed method is compared with traditional energy-based interference detection techniques. The results demonstrate that, once the DTC-based interference analysis model is established, the "Judgment Range of the Distance to the Center" (JR-DTC) can be employed to rapidly assess interference conditions. This approach achieves superior detection performance compared to energy-based methods, even with few sampling points, enabling real-time and accurate analysis of CFI in mega-constellation scenarios.
- The position of the received signal's probability density function on the manifold is mapped into three-dimensional space through affine embedding, providing a visual representation of the interference signals received by Earth stations. This visualization not only enhances the interpretability of the results but also offers an intuitive reference for subsequent link design, system optimization, and frequency planning.

II. Interference Analysis Model

A. Interference Scenario

Consider the communication scenario of two NGSO satellite communication systems that operating at the same frequency

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

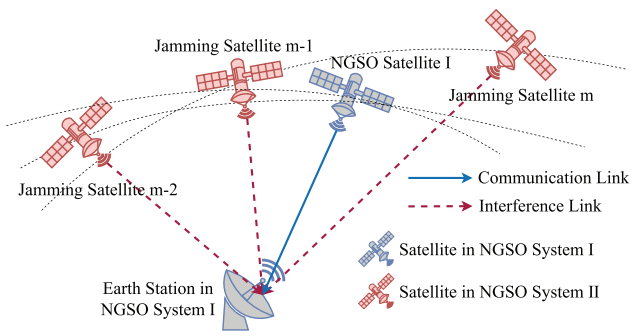


Fig. 1. Interference scenario between NGSO systems.

band, as shown in Fig. 1. NGSO System I is the interfered NGSO satellite communication system, in which Satellite I communicates with the Earth station of this system. NGSO System II is the satellite communication system that causes interference to System I. At a given moment, the number of interfering satellites from NGSO System II visible to the Earth station of NGSO System I is denoted as m. These m satellites have their antenna beams directed toward their respective subsatellite points and may all potentially cause CFI to the communication of the Earth station in NGSO System I.

B. Satellite Signal Model

Assuming that two satellite communication systems, NGSO system I and NGSO system II, both use Amplitude Phase Shift Keying (APSK) modulation [26], and the channel noise is additive white Gaussian noise (AWGN), then the time-domain expression for the interference-free signal received by NGSO system I's Earth station is

$$r_0(t) = A_0 e^{j[2\pi f t + \phi(t)]} + n_0(t), \tag{1}$$

where t is the time, A_0 is the amplitude of the signal, $\phi(t)$ is the phase, $n_0(t)$ represents the AWGN, which is a normally distributed random variable with mean $\mu_0 = 0$ and variance σ_0^2. f is the frequency of the carrier. When m satellites in NGSO System II cause interference, the signal received by NGSO System I's Earth station should be the sum of the interference-free signal and all the interfering signals, denoted as $r_m(t)$, whose time-domain expression is

$$r_m(t) = A_0 e^{j[2\pi f t + \phi(t)]} + n_0(t) + \sum_{i=1}^{m} \left[A_i e^{j[2\pi f t + \phi_i(t)]} + n_i(t) \right], \tag{2}$$

where i denotes the index of the interfering satellite in NGSO system II. A_i and ϕ_i are the signal amplitude and the signal phase of the i-th interfering satellite, respectively. $n_i(t)$ is the AWGN of the i-th interfering satellite that follows a normal distribution with a mean of $\mu_i = 0$ and a variance of σ_i^2.

C. Downlink Transmission Model

Since the CFI analysis process of uplink and downlink in satellite communication is basically the same, this paper takes the downlink as an example to study the CFI between NGSO constellation systems. It is assumed that the location of the Earth station in NGSO system I is fixed, its latitude and longitude coordinates are (θ_G, φ_G), its altitude from the ground is h_G, and its antenna diameter is D_G. The latitude and longitude coordinates of the subsatellite point of NGSO satellite I are $(\theta_{\mathrm{I}}, \varphi_{\mathrm{I}})$, the altitude of the satellite above the ground is h_{I}, the diameter of its antenna and its transmitting power are D_{I} and P_t, respectively. The latitude and longitude coordinates of the subsatellite points of the interfering satellites in NGSO system II are $(\theta_{\mathrm{II}1}, \varphi_{\mathrm{II}1}), (\theta_{\mathrm{II}2}, \varphi_{\mathrm{II}2}), \cdots, (\theta_{\mathrm{II}m}, \varphi_{\mathrm{II}m})$. The altitude of the satellites above the ground are $h_{\mathrm{II}1}, h_{\mathrm{II}2}, \cdots, h_{\mathrm{II}m}$. The diameter of the satellite transmitting antennas and their transmitting power are $D_{\mathrm{II}1}, D_{\mathrm{II}2}, \cdots, D_{\mathrm{II}m}$ and $I_{\mathrm{II}1}, I_{\mathrm{II}2}, \cdots, I_{\mathrm{II}m}$, respectively.

The pattern of all satellites' transmitting antennas and Earth station's receiving antenna refer to ITU-R S.1528 Recommendation and ITU-R S.465 Recommendation, respectively [27], [28]. For LEO satellites satisfying $D_t/\lambda < 35$ and NGSO Earth station satisfying $D_r/\lambda < 50$, The formulas for calculating antenna transmit gain $G_t(\psi_t)$ and antenna receive gain $G_r(\psi_r)$ in the direction of off-axis angle ψ are shown in (3), shown at the bottom of the page, and (4), respectively

$$G_r(\psi_r) = \begin{cases} 32 - 25\log\psi_r, \psi_{\min} \le \psi_r < 48^\circ \\ -10, 48^\circ \le \psi_r \le 180^\circ, \end{cases} \tag{4}$$

where ψ_t and ψ_r represent the off-axis angle of the transmitting antenna of NGSO satellite and the receiving antenna of NGSO system I's Earth station, respectively. $\psi_{\min} = \max(2, 114(D_r/\lambda)^{-1.09})$; $G_{t\max}$, ϕ_b and L_F are the maximum gain of the main lobe, the half power beam width and the far-out side-lobe level of the NGSO satellite's transmitting antenna, respectively. L_F is close to 0 for the ideal antenna radiation model. λ is the signal wavelength; D_t and D_r are the diameter of the NGSO satellite's transmitting antenna and the Earth station's receiving antenna, respectively. According to the calculation formula of the angle between vectors, ψ_t and ψ_r can be expressed as

$$\psi_t = \arccos \frac{\overrightarrow{S_{\mathrm{II}}O} \cdot \overrightarrow{S_{\mathrm{II}}E_{\mathrm{I}}}}{\left|\overrightarrow{S_{\mathrm{II}}O}\right| \left|\overrightarrow{S_{\mathrm{II}}E_{\mathrm{I}}}\right|} \tag{5}$$

$$\psi_r = \arccos \frac{\overrightarrow{E_{\mathrm{I}}S_{\mathrm{I}}} \cdot \overrightarrow{E_{\mathrm{I}}S_{\mathrm{II}}}}{\left|\overrightarrow{E_{\mathrm{I}}S_{\mathrm{I}}}\right| \left|\overrightarrow{E_{\mathrm{I}}S_{\mathrm{II}}}\right|}, \tag{6}$$

where O, S_{I}, S_{II} and E_{I} represent the points corresponding to the origin of geocentric cartesian coordinate system, the satellite in NGSO system I, the satellite in NGSO system

$$G_t(\psi_t) = \begin{cases} G_{t\max} - 3(\psi_t/\psi_b)^2, & \psi_b < \psi_t < 1.5\psi_b \\ 20\log(D_t/\lambda) + 5.65 - 25\log(\psi_t/\psi_b), & 1.5\psi_b < \psi_t < 1.5\psi_b \times 10^{0.04(G_{t\max} - 6.75 - L_F)} \\ L_F, & 1.5\psi_b \times 10^{0.04(G_{t\max} - 6.75 - L_F)} < \psi_t < 180^\circ \end{cases} \tag{3}$$

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

II, and the Earth station in NGSO system I, respectively; $|\overrightarrow{vec}|$ represents the modular value of $\overrightarrow{vec}$; $\overrightarrow{S_{\mathrm{II}}O}$, $\overrightarrow{E_{\mathrm{I}}S_{\mathrm{I}}}$, $\overrightarrow{S_{\mathrm{II}}E_{\mathrm{I}}}$ and $\overrightarrow{E_{\mathrm{I}}S_{\mathrm{II}}}$ are vectors from NGSO system II's satellite to the origin, vectors from NGSO System I's Earth station to NGSO system I's satellite, vectors from NGSO system II's satellite to NGSO system I's Earth station and vectors from NGSO system I's Earth station to NGSO System II's satellite, respectively. According to the conversion relationship between the geodetic coordinate system and the geocentric cartesian coordinate system, their calculation formulas are as follows

$$\overrightarrow{S_{\mathrm{II}}O} = \begin{pmatrix} -(N_e + h_{\mathrm{II}})\cos\varphi_{\mathrm{II}}\cos\theta_{\mathrm{II}}, \\ -(N_e + h_{\mathrm{II}})\cos\varphi_{\mathrm{II}}\sin\theta_{\mathrm{II}}, \\ -\left(N_e\left(1-e^2\right) + h_{\mathrm{II}}\right)\sin\varphi_{\mathrm{II}} \end{pmatrix} \tag{7}$$

$$\overrightarrow{E_{\mathrm{I}}S_{\mathrm{I}}} = \begin{pmatrix} (N_e + h_{\mathrm{I}})\cos\varphi_{\mathrm{I}}\cos\theta_{\mathrm{I}} - (N_e + h_G)\cos\varphi_G\cos\theta_G, \\ (N_e + h_{\mathrm{I}})\cos\varphi_{\mathrm{I}}\sin\theta_{\mathrm{I}} - (N_e + h_G)\cos\varphi_G\sin\theta_G, \\ \left(N_e\left(1-e^2\right) + h_{\mathrm{I}}\right)\sin\varphi_{\mathrm{I}} - \left(N_e\left(1-e^2\right) + h_G\right)\sin\varphi_G \end{pmatrix} \tag{8}$$

$$\overrightarrow{S_{\mathrm{II}}E_{\mathrm{I}}} = -\overrightarrow{E_{\mathrm{I}}S_{\mathrm{II}}} = \begin{pmatrix} (N_e + h_G)\cos\varphi_G\cos\theta_G - (N_e + h_{\mathrm{II}})\cos\varphi_{\mathrm{II}}\cos\theta_{\mathrm{II}}, \\ (N_e + h_G)\cos\varphi_G\sin\theta_G - (N_e + h_{\mathrm{II}})\cos\varphi_{\mathrm{II}}\sin\theta_{\mathrm{II}}, \\ \left(N_e\left(1-e^2\right) + h_G\right)\sin\varphi_G - \left(N_e\left(1-e^2\right) + h_{\mathrm{II}}\right)\sin\varphi_{\mathrm{II}} \end{pmatrix}, \tag{9}$$

where N_e is the prime unitary circle radius, and $N_e = a/\sqrt{1-e^2\sin^2 b}$. $e = \sqrt{(a^2-b^2)/a^2}$ is the eccentricity of the Earth. a and b are the length of the semi-major axis and the length of the semi-minor axis of the Earth ellipsoid, respectively. Since N_e and e are constant, the off-axis angle ψ is a variable that only related to the latitude and longitude of the Earth station and the satellite's subsatellite points, as well as the respective altitude of the Earth station and satellite, that is $\psi(\theta_G, \varphi_G, h_G, \theta_S, \varphi_S, h_S)$, and $(\theta_S, \varphi_S, h_S)$ are the longitude and latitude of the satellite's sub-satellite point and the altitude of the satellite, respectively.

Assuming that signal transmission is subject to free space loss and attenuation caused by clouds, fog, and rain, the formula for calculating the link loss is

$$L = L_{FS} + L_{AR} + L_{AC}. \tag{10}$$

Among them, L_{FS} represents the free space loss, L_{AR} denotes the attenuation caused by rainfall, and L_{AC} indicates the attenuation caused by clouds and fog. Their calculation formulas are as follows

$$L_{FS} = 20\log(4\pi d/\lambda) = 32.4 + 20\log f + 20\log d \tag{11}$$

$$L_{AR} = \gamma_R L_E \tag{12}$$

$$L_{AC} = \frac{L_{red}K_l(f, 273.15)}{\sin\varphi_{Ele}} \quad \text{(dB)} \quad \text{for} \quad 90^\circ \geq \varphi_{Ele} \geq 5^\circ, \tag{13}$$

where f MHz is the signal frequency; d km represents the distance of the communication link between the satellite and the Earth station. For NGSO satellite I and satellites in NGSO system II, its values are the modulus of vector $\overrightarrow{S_{\mathrm{I}}E_{\mathrm{I}}}$ and $\overrightarrow{S_{\mathrm{II}}E_{\mathrm{I}}}$, respectively. γ_R is the specific attenuation, and L_E is the effective path length. Their calculation formulas can be referred to in the ITU-R P. 838 Recommendation [29]. L_{red} is the total columnar content of liquid water (kg/m^2) reduced to a temperature of 273.15 K, exceeded for the probability levels of p. K_l is the cloud liquid water specific attenuation coefficient [(dB/km)/(g/m^3)], and φ_{Ele} is the elevation angle. Their calculation formulas are provided in the ITU-R P. 840-7 Recommendation [30].

Since the signal frequency f is a constant, and γ_R, L_{red}, and K_l can be considered as fixed constants over a short period of time, the link loss depends only on the distance d, the effective path length L_E and the elevation angle φ_{Ele}. Meanwhile, these three variables are determined solely by the latitude and longitude of the Earth station and the satellite's sub-satellite point, as well as the respective altitudes of the Earth station and the satellite. Therefore, the link loss can be expressed as a function of the longitude and latitude of the Earth station and the satellite's sub-satellite point, as well as the respective altitudes of them, i.e., $L(\theta_G, \varphi_G, h_G, \theta_S, \varphi_S, h_S)$. Consequently, the power P_r received by NGSO System I's Earth station from NGSO Satellite I and the aggregate interference power from NGSO System II's satellites can be expressed as

$$P_r(\theta_G, \varphi_G, h_G, \theta_{\mathrm{I}}, \varphi_{\mathrm{I}}, h_{\mathrm{I}}) = P_t + G_{t\mathrm{I}\max} + G_{r\max} - L \tag{14}$$

$$I_r(\theta_G, \varphi_G, h_G, \theta_{\mathrm{II}}, \varphi_{\mathrm{II}}, h_{\mathrm{II}}) = \sum_{i=1}^{m}\left[I_{t\mathrm{II}i} + G_{\mathrm{II}i}(\psi_{t\mathrm{II}i}) + G_{r\mathrm{II}i}(\psi_{r\mathrm{II}i}) - L_{\mathrm{II}i}\right], \tag{15}$$

where $L_{\mathrm{II}i}$ represents the link loss of the i-th satellites in NGSO system II.

D. Parametric Representation Model of Statistical Manifolds

Based on information geometry theory, the mathematical model of the signals received by NGSO system I's Earth station are constructed in this subsection. Assume that the interference-free signal $r_0(t)$ received by NGSO system I's Earth station are independently sampled for N times, and a $M \times N$ matrix can be obtained by sampling M points each time [34]

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N] = \begin{bmatrix} x_1(1) & x_2(1) & \cdots & x_N(1) \\ x_1(2) & x_2(2) & \cdots & x_N(2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(M) & x_2(M) & \cdots & x_N(M) \end{bmatrix}, \tag{16}$$

where the M-dimensional vector $\mathbf{x}_i = [x_i(1), x_i(2), \cdots, x_i(M)]^T$ is the data obtained from the i-th ($1 \leq i \leq N$) sampling and follows the zero-mean Gaussian distribution, and its probability density function can be expressed as

$$p(\mathbf{x}|\mathbf{R_X}; \theta_{\mathrm{I}}, \varphi_{\mathrm{I}}) = \frac{\exp\left[-\frac{1}{2}\mathbf{x}^T\mathbf{R}_{\mathbf{X}}^{-1}(\theta_{\mathrm{I}}, \varphi_{\mathrm{I}})\mathbf{x}\right]}{\sqrt{(2\pi)^N \det\mathbf{R_X}(\theta_{\mathrm{I}}, \varphi_{\mathrm{I}})}}. \tag{17}$$

In the above formula, $\mathbf{R_X}(\theta_{\mathrm{I}}, \varphi_{\mathrm{I}}) \in \mathbf{C}^{N\times N}$ is the covariance matrix composed of all the sampled data, which can be calculated by (18), shown at the bottom of the next page.

In (18), $\mathbf{x}(i)$ represents column i ($1 \leq i \leq N$) in the sample matrix $\mathbf{X}$. According to the information geometry

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

theory, a differentiable manifold structure is formed from the probability distribution family parameterized by $\mathbf{R_X}$, which is called statistical manifold [31]. Since the parameter $\mathbf{R_X}$ of the manifold is a covariance matrix, the manifold can also be called a matrix manifold, denoted as

$$\mathcal{M}_{\mathbf{X}} = \left\{ p\left(\mathbf{x} \mid \mathbf{R_x}; \theta_{\mathrm{I}}, \varphi_{\mathrm{I}}\right) \mid \theta_{\mathrm{I}} \in [-\pi, \pi), \varphi_{\mathrm{I}} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \right\}. \quad (19)$$

Similarly, when m satellites of NGSO system II interfere with the Earth station of NGSO system I, the signal $r_m(t)$ received by NGSO system I's Earth station is independently sampled N times, and M points are collected each time to obtain a matrix of $M \times N$

$$\mathbf{Y} = \left[\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_N\right] = \begin{bmatrix} y_1(1) & y_2(1) & \cdots & y_N(1) \\ y_1(2) & y_2(2) & \cdots & y_N(2) \\ \vdots & \vdots & \ddots & \vdots \\ y_1(M) & y_2(M) & \cdots & y_N(M) \end{bmatrix}, \quad (20)$$

where the M-dimensional vector $\mathbf{y}_i = \left[y_i(1), y_i(2), \cdots, y_i(M)\right]^T$ is the data obtained from the i-th $(1 \le i \le N)$ sampling and follows the zero-mean Gaussian distribution, and its probability density function can be expressed as

$$p(\mathbf{y}|\mathbf{R_Y}; \theta_{\mathrm{II}}, \varphi_{\mathrm{II}}) = \frac{\exp\left[-\frac{1}{2}\mathbf{y}^T \mathbf{R}_{\mathbf{Y}}^{-1}\left(\theta_{\mathrm{II}}, \varphi_{\mathrm{II}}\right)\mathbf{y}\right]}{\sqrt{(2\pi)^N \det \mathbf{R_Y}\left(\theta_{\mathrm{II}}, \varphi_{\mathrm{II}}\right)}}. \quad (21)$$

In (21), the calculation formula of the covariance matrix $\mathbf{R_Y}\left(\theta_{\mathrm{II}}, \varphi_{\mathrm{II}}\right) \in \mathbf{C}^{N \times N}$ is as (22), shown at the bottom of the page, where $\mathbf{y}(i)$ represents column i $(1 \le i \le N)$ in the sample matrix $\mathbf{Y}$. According to the information geometry theory, a family of probability distributions parameterized by $\mathbf{R_Y}$ can also form a matrix manifold [31], denoted as

$$\mathcal{M}_{\mathbf{Y}} = \left\{ p(\mathbf{y}|\mathbf{R_Y}; \theta_{\mathrm{II}}, \varphi_{\mathrm{II}}) | \theta_{\mathrm{II}} \in [-\pi, \pi), \varphi_{\mathrm{II}} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \right\}. \quad (23)$$

III. Interference Analysis Method Based on Manifold DTC

Since $R_{\mathbf{X}}$ and $R_{\mathbf{Y}}$ are different, the matrix manifolds $\mathcal{M}_{\mathbf{X}}$ and $\mathcal{M}_{\mathbf{Y}}$ correspond to two different submanifolds on a multidimensional Gaussian distribution matrix manifold, respectively. Therefore, this section first uses the Monte Carlo simulation method to determine the "center" of the probability distribution function (PDF) on the matrix manifold. This PDF describes the sampling data of the interference-free signal received by the Earth station of NGSO system I, and the "center" corresponds to the "center matrix" of the submanifold $\mathcal{M}_{\mathbf{X}}$. This "center" matrix is utilized as the "initial reference point" for subsequent interference analysis. Then, the distance between the covariance matrix of the sampling data and the "center" matrix in each Monte Carlo simulation is calculated, that is, the DTC. And subsequently, the threshold value of DTC is obtained by setting the false alarm rate. When the DTC of the covariance matrix obtained from the sampling data is greater than the threshold value, it is considered that the covariance matrix of the sampling data is significantly different from that obtained from the interference-free signal. This means that the corresponding sampling data is interfered with. For the scenario of potential interference from multiple NGSO system II's satellites, the upper and lower bounds of the DTC, i.e. the "judgment range of distance to center" (JR-DTC), under different interference conditions are obtained through Monte Carlo method. By analyzing the DTC of the covariance matrix of the sampling data falls in which JR-DTC, a conclusion about the interference situation at this time is drawn. Finally, by affine embedding method, the potential function of the statistical distribution of the signal sampling data is associated with the latitude and longitude of the subsatellite points of NGSO system II's satellites in three-dimensional space, and the signals received by the NGSO system I's Earth station are visualized, which realizes the analysis and visual representation of the variation law of CFI between NGSO systems.

The main steps of the above mentioned downlink CFI analysis method between NGSO systems based on information geometry are shown in Fig. 2.

A. The Calculation of "Center" Matrix

For the hypothesis testing problem with unknown parameters, the detection model corresponds to two uncertain regions on the corresponding matrix manifold, which can be obtained by the prior distribution of the parameters. Therefore, in order to analyze the downlink CFI between NGSO systems, it is necessary to first determine the "center" of the uncertain region on the manifold. This uncertain region corresponds to the PDF of the covariance matrix of the interference-free signal received by the NGSO system I's Earth station. Since the "center" of the uncertain region on a matrix manifold cannot be calculated through analytical expression, the Monte Carlo

$$\mathbf{R_X} = \frac{1}{M-1}\begin{bmatrix} \mathrm{cov}\left(\mathbf{x}(1), \mathbf{x}(1)\right) & \mathrm{cov}\left(\mathbf{x}(1), \mathbf{x}(2)\right) & \cdots & \mathrm{cov}\left(\mathbf{x}(1), \mathbf{x}(N)\right) \\ \mathrm{cov}\left(\mathbf{x}(2), \mathbf{x}(1)\right) & \mathrm{cov}\left(\mathbf{x}(2), \mathbf{x}(2)\right) & \cdots & \mathrm{cov}\left(\mathbf{x}(2), \mathbf{x}(N)\right) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{cov}\left(\mathbf{x}(N), \mathbf{x}(1)\right) & \mathrm{cov}\left(\mathbf{x}(N), \mathbf{x}(2)\right) & \cdots & \mathrm{cov}\left(\mathbf{x}(N), \mathbf{x}(N)\right) \end{bmatrix} \quad (18)$$

$$\mathbf{R_Y} = \frac{1}{M-1}\begin{bmatrix} \mathrm{cov}\left(\mathbf{y}(1), \mathbf{y}(1)\right) & \mathrm{cov}\left(\mathbf{y}(1), \mathbf{y}(2)\right) & \cdots & \mathrm{cov}\left(\mathbf{y}(1), \mathbf{y}(N)\right) \\ \mathrm{cov}\left(\mathbf{y}(2), \mathbf{y}(1)\right) & \mathrm{cov}\left(\mathbf{y}(2), \mathbf{y}(2)\right) & \cdots & \mathrm{cov}\left(\mathbf{y}(2), \mathbf{y}(N)\right) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{cov}\left(\mathbf{y}(N), \mathbf{y}(1)\right) & \mathrm{cov}\left(\mathbf{y}(N), \mathbf{y}(2)\right) & \cdots & \mathrm{cov}\left(\mathbf{y}(N), \mathbf{y}(N)\right) \end{bmatrix} \quad (22)$$

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

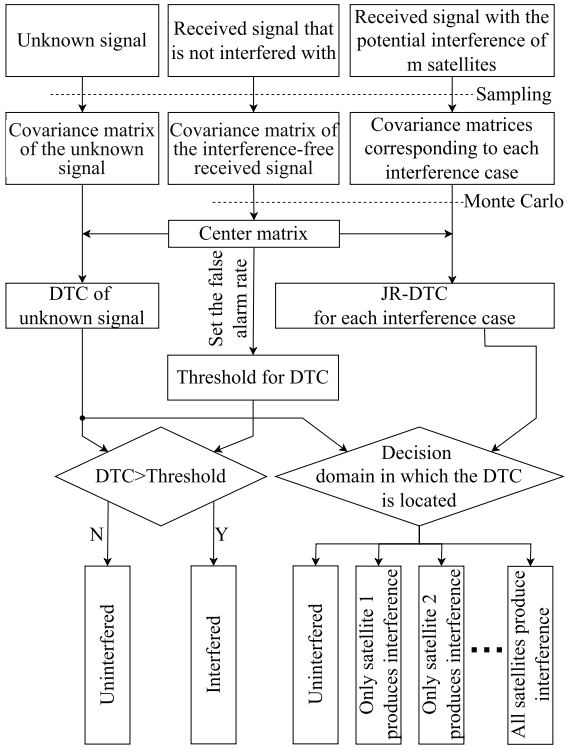


Fig. 2. Flowchart of the downlink CFI analysis method between NGSO systems based on information geometry.

method is used to repeatedly sample the received signals for multiple times in order to find the "center" of the uncertain region.

There are many ways to measure the "distance" or difference between two points on a matrix manifold, including the Affine Invariant Riemannian Metric (AIRM), Symmetric Kullback-Leibler Divergence (SKLD), Log-Euclidean Metric (LEM), and Jansen-Bregman LogDet Divergence (JBLD). Different metrics reflect different manifold geometric structure and differences between two points. AIRM is the Riemannian metric that reflects the true geodesic distance between two points on a Riemannian manifold and has affine invariance; SKLD is a measure of the difference between different probability distributions with symmetric properties; LEM is defined by combining the logarithm of symmetric positive definite matrices and Euclidean distance, but does not fully reflect the geometric relationship (i.e., geodesic distance) between two points on the Riemannian manifold; JBLD is based on the concepts of matrix determinant and logarithmic determinant to measure the difference between two covariance matrices, but does not fully reflect the true geodesic distance on the Riemannian manifold. Therefore, in this paper, the interference analysis method based on DTC takes two common metrics as examples, namely AIRM and SKLD.

Under AIRM metric, assume $\mathbf{R}_{\mathbf{X}k}$ is the covariance matrix obtained in the k-th Monte Carlo simulation, then "center" matrix $\overline{\mathbf{R}}_{\mathbf{X}A}$ can be obtained through the following iteration [32]

$$\overline{\mathbf{R}}_{\mathbf{X}A(i+1)} = \overline{\mathbf{R}}_{\mathbf{X}Ai}^{1/2}\exp\left(\frac{\tau}{Q}\sum_{k=1}^{Q}\log\left(\overline{\mathbf{R}}_{\mathbf{X}Ai}^{-1/2}\mathbf{R}_{\mathbf{X}k}\overline{\mathbf{R}}_{\mathbf{X}Ai}^{-1/2}\right)\right)\overline{\mathbf{R}}_{\mathbf{X}Ai}^{1/2}, \tag{24}$$

where $\overline{\mathbf{R}}_{\mathbf{X}Ai}$ and $\overline{\mathbf{R}}_{\mathbf{X}A(i+1)}$ are the "center" matrix obtained after the i-th and $(i+1)$-th iterations respectively, $\tau(0 \le \tau \le 1)$ is the iteration step size or learning rate, Q is the number of Monte Carlo simulation, $\exp(\cdot)$ and $\log(\cdot)$ represent the matrix exponential and matrix logarithm, respectively.

Under SKLD metrics, the "center" matrix $\overline{\mathbf{R}}_{\mathbf{X}S}$ can be directly calculated by (25), shown at the bottom of the page, [33].

B. The Calculation of DTC and Its Threshold

For the covariance matrix $\mathbf{R}_{\mathbf{X}1}, \mathbf{R}_{\mathbf{X}2}, \cdots, \mathbf{R}_{\mathbf{X}Q}$ obtained in Q Monte Carlo simulations, the distance between them and the "center" matrix $\bar{\mathbf{R}}_{\mathbf{X}A}$, namely DTC, is calculated respectively under the AIRM metric by the following formula

$$D_A\left(\bar{\mathbf{R}}_{\mathbf{X}A}, \mathbf{R}_{\mathbf{X}k}\right) = \left\|\log\left(\mathbf{R}_{\mathbf{X}k}^{1/2}\bar{\mathbf{R}}_{\mathbf{X}A}^{-1}\mathbf{R}_{\mathbf{X}k}^{1/2}\right)\right\|_F = \sum_{i=1}^{M}\ln\left(\Lambda_i\right), \tag{26}$$

where Λ_i is the i-th eigenvalue of the matrix $\mathbf{R}_{\mathbf{X}k}^{1/2}\bar{\mathbf{R}}_{\mathbf{X}A}^{-1}\mathbf{R}_{\mathbf{X}k}^{1/2}$, and $\|\cdot\|_F$ represents the Frobenius norm of the matrix. Under SKLD metric, the DTC of the covariance matrix $\mathbf{R}_{\mathbf{X}1}, \mathbf{R}_{\mathbf{X}2}, \cdots, \mathbf{R}_{\mathbf{X}Q}$ is calculated as follows

$$D_S\left(\bar{\mathbf{R}}_{\mathbf{X}S}, \mathbf{R}_{\mathbf{X}k}\right) = \frac{1}{2}\mathrm{tr}\left(\mathbf{R}_{\mathbf{X}k}^{-1}\bar{\mathbf{R}}_{\mathbf{X}S} + \bar{\mathbf{R}}_{\mathbf{X}S}^{-1}\mathbf{R}_{\mathbf{X}k} - 2\mathbf{I}\right), \tag{27}$$

where $\mathrm{tr}(\cdot)$ represents the trace of matrix $\mathbf{R}_{\mathbf{X}k}^{-1}\bar{\mathbf{R}}_{\mathbf{X}S} + \bar{\mathbf{R}}_{\mathbf{X}S}^{-1}\mathbf{R}_{\mathbf{X}k} - 2\mathbf{I}$ and $\mathbf{I}$ is the identity matrix. The Q DTC values of AIRM and SKLD metrics can then be obtained respectively. Arrange the Q DTC values in the order from small to large, set the false alarm rate to P_f, then the DTC threshold D_T is the value corresponding to the $Q(1-P_f)$-th DTC. When the DTC of the covariance matrix $\mathbf{R}_{\mathbf{Y}}$ of a sampled data is greater than the threshold value D_T, it is considered that the covariance matrix of the sampled data is very different from that of the sampled data of interference-free signal. This means that the sampled data is interfered with. The analysis process can be expressed as

$$\mathrm{D}\left(\bar{\mathbf{R}}_{\mathbf{X}}, \mathbf{R}_{\mathbf{Y}}\right) \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} D_T, \tag{28}$$

where $\mathcal{H}_0$ indicates that the signal received by the NGSO system I's Earth station is not interfered with, and $\mathcal{H}_1$ indicates that the signal received by the NGSO system I's Earth station is interfered with.

$$\overline{\mathbf{R}}_{\mathbf{X}S} = \left(\sum_{i=1}^{Q}\mathbf{R}_{\mathbf{X}i}^{-1}\right)^{\frac{-1}{2}}\left[\left(\sum_{i=1}^{Q}\mathbf{R}_{\mathbf{X}i}^{-1}\right)^{\frac{1}{2}}\sum_{i=1}^{Q}\mathbf{R}_{\mathbf{X}i}\left(\sum_{i=1}^{Q}\mathbf{R}_{\mathbf{X}i}^{-1}\right)^{\frac{1}{2}}\right]^{\frac{1}{2}}\left(\sum_{i=1}^{Q}\mathbf{R}_{\mathbf{X}i}^{-1}\right)^{\frac{-1}{2}} \tag{25}$$

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

C. Calculation of the JR-DTC for Each Interference Case

In addition to the detection of whether the signal received by NGSO system I's Earth station is interfered with, we hope to further analyze the interference situation when there is interference, that is, which satellites have caused interference to the communication of NGSO system I. When the parameters of the potential interfering satellite are known, we can use the "center" matrix and Monte Carlo method to set the JR-DTC for each interference case, and then analyze the current interference situation according to the DTC of the covariance matrix of the unknown signal's sampling data.

When there are m satellites that have the potential to cause interference to NGSO system I, there are a total of 2^m interference cases, that is, all satellites do not cause interference, only satellite 1 causes interference, only satellite 2 causes interference, $\cdots$, all satellites cause interference, denoted as Case0, Case1, $\cdots$, Case $2^m - 1$. Q Monte Carlo simulations are carried out under each interference condition for the signals received by NGSO system I's Earth station, and Q individual covariance matrices are obtained. Then, the DTC of Q individual covariance matrices in each interference case is calculated respectively, and the upper and lower bounds of DTC in each interference case are obtained, that is, the JR-DTC. For covariance matrix $\mathbf{R_Y}$ of an unknown signal's sampling data, by analyzing which JR-DTC its DTC falls in, we can get the conclusion of what kind of interference situation is at this time. The analysis process can be expressed as

$$\mathrm{D}\left(\bar{\mathbf{R}}_{\mathbf{X}}, \mathbf{R_Y}\right) \in \begin{cases} [D_{0d}, D_{0u}], & \text{Case0} \\ [D_{1d}, D_{1u}], & \text{Case1} \\ \vdots & \vdots \\ \left[D_{(2^m-1)d}, D_{(2^m-1)u}\right], & \text{Case}2^m - 1 \end{cases} \tag{29}$$

where $D_{(2^m-1)d}$ represents the lower bound of the DTC value for interference situation Case$2^m - 1$, $D_{(2^m-1)u}$ represents the upper bound of the DTC value for interference situation Case$2^m - 1$.

Eqs. (28) and (29) are the two most important steps of the method proposed in this paper, where (28) illustrates the process of determining whether the received signal is interfered with by DTC and the interference judgment threshold; (29) illustrates the process of determining the interference conditions of the received signal by utilizing the DTC and the interference judgment domain.

D. Affine Embedding Representation of Interference Cases

Since the PDF of any exponential distribution family corresponds to a unique potential function, and any potential function corresponds to a unique PDF of exponential distribution family, and the Gaussian distribution is a member of the exponential family of distributions, therefore, it is possible to use the potential function of the distribution to represent the multidimensional Gaussian manifolds in R^{n+1}-dimensional Euclidean space by affine embedding [35].

According to the potential function formula corresponding to the multidimensional Gaussian distribution, when the mean value is 0, the potential function Φ can be represented by the local parameter $(\boldsymbol{\mu}, \mathbf{R})$ as

$$\Phi(\boldsymbol{\mu}, \mathbf{R}) = \Phi(0, \mathbf{R}) = \frac{1}{2}\ln\det \mathbf{R_Y} + \frac{N}{2}\ln 2\pi. \tag{30}$$

In the above formula, N is the dimension of the covariance matrix. Because the value of the covariance matrix $\mathbf{R_Y}$ of the sampled data is only related to the latitude and longitude $(\theta_{\mathrm{II}}, \varphi_{\mathrm{II}})$ and the orbital altitude h_{II} of the satellite, and for the same satellite, its orbital altitude can be approximated to a fixed value. Therefore, the value of the potential function Φ can be expressed as a binary function of the latitude and longitude $(\theta_{\mathrm{II}}, \varphi_{\mathrm{II}})$ of the subsatellite point. With θ_{II} and φ_{II} as horizontal and vertical axes and the value of potential function as Z-axis coordinates, a three-dimensional coordinate system can be established to obtain a three-dimensional surface diagram of "latitude-longitude-potential function", so as to realize the visualization of the interference between NGSO systems.

IV. Simulation and Verification

Applying the mathematical methods and models mentioned above and taking the downlink CFI analysis between "OneWeb" constellation and "Starlink" constellation as an example, simulation verification is carried out. In simulation, we consider the realistic satellite parameter and orbital data of "OneWeb" and "Starlink". Meanwhile, the communication frequency band of the "OneWeb" constellation system and "Starlink" constellation system are thought to be completely overlapping. The possible interference caused by two satellites in "Starlink" constellation to the downlink between a satellite and the Earth station in "OneWeb" system is analyzed. The Earth model is WGS-84 and the coordinate system used in the simulation is the Earth Centered Fixed Coordinate System (ECF). The efficiency of the antenna is 100%, and the influence of antenna polarization and the modulation method is ignored. Other simulation parameters are shown in TABLE. I.

Set the number of Monte Carlo simulations to 1000 times, and samples the signal 10 times independently in each Monte Carlo simulation, and set the number of sampling points for each independent sample to 15 ~ 50, 60, 70, 80, 90, 100. The influence of different sampling points on the performance of the interference analysis method based on manifold DTC is analyzed.

A. Simulation of DTC and Its Threshold Value Under AIRM Metric

First, the parameters in TABLE. I are taken into (14) to obtain the interference-free signal power of the "OneWeb" satellite received by the "OneWeb" Earth station, and the amplitude of the received signal is calculated through the power. Then, the amplitude is brought into the model of the interference-free signal in (1), and 10 independent samples are taken under each number of sampling points, and the data of 10 independent samples is brought into (18), thus obtaining a covariance matrix with a size of 10×10 that corresponds to a point on the manifold $\mathcal{M}_{\mathbf{X}}$ constructed in (19). Then, 1000 Monte Carlo simulations are performed for the above steps to obtain 1000 covariance matrices, and the

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SIMULATION PARAMETERS

Parameters	Values
Transmission power of "OneWeb" satellite	8.5 dBW
Transmitting antenna peak gain of "OneWeb" satellite	25.9 dBi
Latitude and longitude of the subsatellite point of "OneWeb" satellite	$(42.635^\circ N, 114.421^\circ E)$
Altitude of "OneWeb" satellite	1202.01 km
"OneWeb" satellite antenna diameter	1 m
Half power beam width of "OneWeb" satellite antenna	3.2°
Receiving antenna peak gain of "OneWeb" Earth station	32 dBi
Receiving antenna diameter of "OneWeb" Earth station	1.2 m
Latitude and longitude of "OneWeb" Earth station	(43.12°N,115.73°E)
Receiving antenna pattern of "OneWeb" Earth station	ITU-R S.465-6
Transmission power of jamming satellite 1 in "Starlink"	8 dBW
Transmitting antenna peak gain of jamming satellite 1 in "Starlink"	44 dBi
Transmitting antenna half power beam angle of jamming satellite 1 in "Starlink"	8.9°
Transmitting antenna diameter of jamming satellite 1 in "Starlink"	0.6 m
Latitude and longitude of the subsatellite point of jamming satellite 1 in "Starlink"	$(43.509^\circ N, 116.851^\circ E)$
Altitude of jamming satellite 1 in "Starlink"	479.3 km
Transmission power of jamming satellite 2 in "Starlink"	8 dBW
Transmitting antenna peak gain of jamming satellite 2 in "Starlink"	34 dBi
Transmitting antenna half power beam angle of jamming satellite 2 in "Starlink"	8.9°
Transmitting antenna diameter of jamming satellite 2 in "Starlink"	0.6 m
Latitude and longitude of the subsatellite point of jamming satellite 2 in "Starlink"	$(41.905^\circ N, 115.589^\circ E)$
Altitude of jamming satellite 2 in "Starlink"	492.3 km
Antenna pattern of "OneWeb" and "Starlink" satellite	ITU-R S.1528 1.2 Circular
Propagation speed of electromagnetic wave	3×10^8 m/s
Major semi-axis radius of the Earth	6378136.49 m
Minor semi-axis radius of the Earth	6356755.00 m
Communication frequency	11.51 GHz
Communication bandwidth	32 MHz
Noise temperature of the receiver	290 K
Attenuation model for rain	ITU-R P.618-12
Outage rate in rain attenuation model	0.1
Attenuation model due to clouds and fog	ITU-R P.840-7
Cloud ceiling	3 km
Cloud layer thickness	0.5 km
Cloud temperature	273.15 K
Liquid water content density value	0.0001 kg/m^3
False alarm rate	0.03

1000 covariance matrices are brought into (24) for iteration to obtain the "center" matrix of the manifold $\mathcal{M}_{\mathbf{X}}$. In order to observe the difference between the mean matrix obtained after each iteration and the "center" matrix of the manifold $\mathcal{M}_{\mathbf{X}}$, the average DTC between the 1000 covariance matrices and the obtained mean matrix is calculated after each iteration. Since the Riemann mean matrix under the AIRM metric needs to be solved by iterative method, the iteration number under the AIRM metric is set to 30, and the iteration step is 0.1. Since the variation curve of average DTC with the number of iterations is basically the same under different sampling points, this paper only shows the curve of average DTC when the sampling points are 15 and 100, as shown in Fig. 3.

As can be seen from Fig. 3, under the AIRM metric, the average DTC of the sampled data's covariance matrix obtained in each Monte Carlo simulation gradually decreases with the increase of the number of iterations of the "center" matrix, indicating that after each iteration, the mean matrix is closer to the actual "center" on the matrix manifold $\mathcal{M}_{\mathbf{X}}$. When the number of iterations exceeds 20, the decreasing trend of the average DTC basically remains unchanged, which indicates that the mean matrix at this time is very close to the actual "center" of the matrix manifold $\mathcal{M}_{\mathbf{X}}$. At the same time, comparing the average DTC under the two cases of 15 and 100 sampling points, it is observed that under the case of the same number of iterations, the more sampling points, the smaller the average DTC, because the more sampling points, the more accurate the probability distribution of the sampled data, and the closer they are to the "center" of $\mathcal{M}_{\mathbf{X}}$.

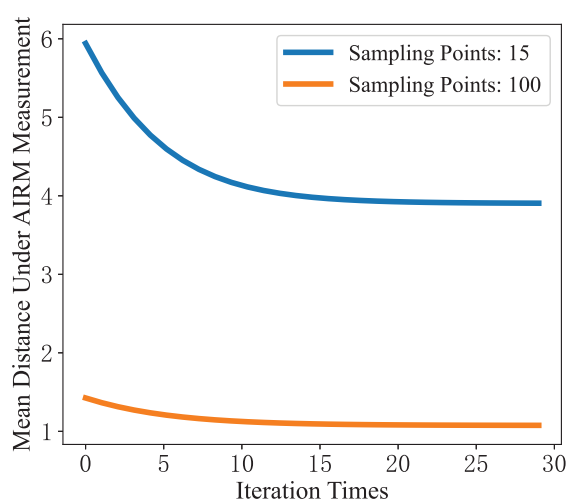
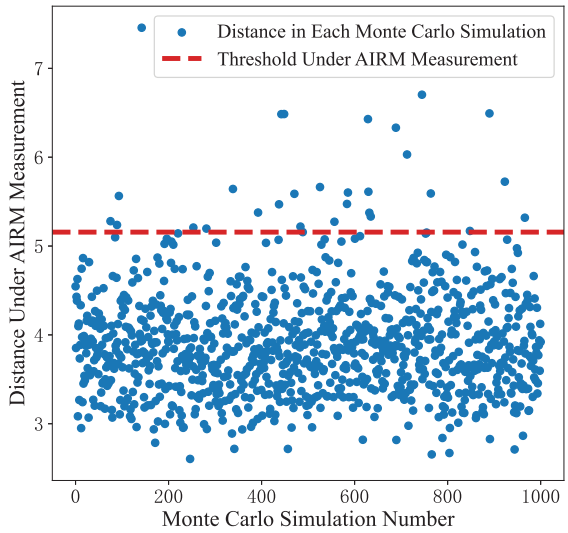
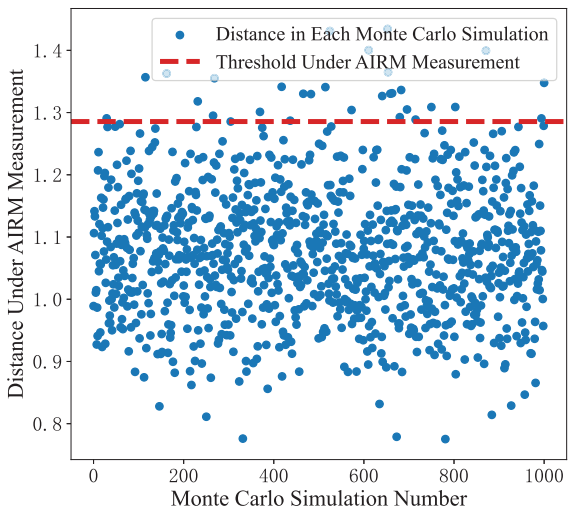


Fig. 3. The change of average DTC with the number of iterations under the AIRM metric.

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.



(a) The number of sampling points is 15.



(b) The number of sampling points is 100.

Fig. 4. The DTC and threshold obtained under the AIRM metric.

Based on the above analysis of the relationship between the number of iterations and the average DTC, and taking into account both the accuracy of the calculation results and the time consumed by the computation, the number of iterations for calculating the "center" matrix under the AIRM metric is set to 30. On this basis, all the DTCs obtained by Monte Carlo simulation under each sampling number is sorted from smallest to largest, and the false alarm rate P_f is set to 0.03, then the value of $Q(1 - P_f)$ is 970, that is, the threshold is the value corresponding to the 970th DTC in the sorted DTC. Taking the Monte Carlo simulation index as the horizontal axis and the DTC under the AIRM metric obtained in each Monte Carlo simulation as the vertical axis, scatter plots were drawn for sampling points of 15 and 100. Additionally, a red horizontal line is added to the scatter plots to indicate the threshold value, resulting in Fig. 4.

From Fig. 4, it is observed that when the sampling point is 15, the value of DTC is mainly distributed between 3 and 5, the range is 2, the variance is 0.35 and the threshold value is 5.16. While when the sampling point is 100, the value of DTC is mainly distributed between 0.9 and 1.3, the range is 0.4, the variance is 0.01 and the threshold value is 1.29. Therefore, it can be found that the numerical distribution of the DTC becomes dense as the number of sampling points increases. At the same time, because the distribution of DTC is denser, the value of the threshold decreases gradually with the increase of the number of sampling points.

B. Simulation of DTC and its Threshold Value Under SKLD Metric

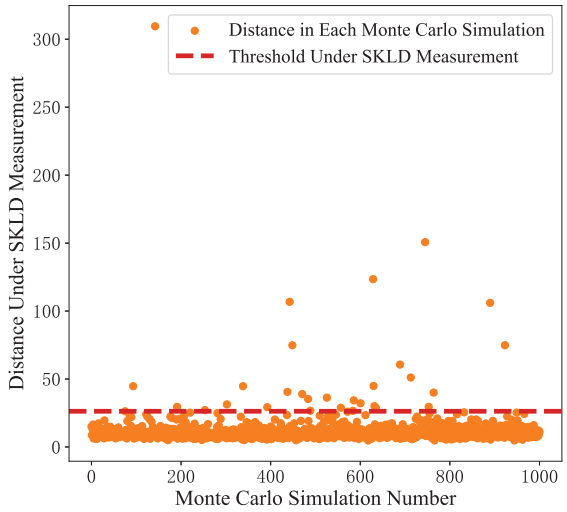
The covariance matrix obtained in Section IV. A is brought into (25) to obtain the "center" matrix under the SKLD metric. Then, the DTC between each covariance matrix and the "center" matrix is calculated and arranged in order from smallest to largest. Similarly, if the false alarm rate P_f is set to 0.03, the value of $Q(1 - P_f)$ is 970, that is, the value corresponding to the 970th sorted DTC is selected as the threshold value under SKLD metric, the Monte Carlo simulation index is taken as the horizontal axis, and the value of DTC under the SKLD metric obtained in each Monte Carlo simulation is taken as the vertical axis to draw scatter plots with sampling points of 15 and 100. A red horizontal line on the scatter plots represents the threshold value, as shown in Fig. 5.

From Fig. 5, it is observed that when the sampling point is 15, the value of DTC is mainly distributed between 0 and 20, the range is 20, the variance is 175.91 and the threshold value is 26.27. While when the sampling point is 100, the value of DTC is mainly distributed between 0.4 and 0.8, the range is 0.4, the variance is 0.014 and the threshold value is 0.85. By comparing Fig. 5 and Fig. 4, it can be observed that the variation trend of the density of the DTC value distribution under the SKLD metric with the number of sampling points is basically the same as that under the AIRM metric. The difference lies in the fact that regardless of whether the number of sampling points is 15 or 100, the distribution range of the DTC value under the SKLD metric is wider than that under the AIRM metric. From this, it can be observed that although the interference analysis method based on manifold DTC under SKLD metric proposed in this paper has the advantage of not requiring iteration to obtain the "center" matrix, the distribution of its DTC value is sparser under the same number of sampling points, and there is a shortcoming of relative larger error in the numerical calculation of DTC.

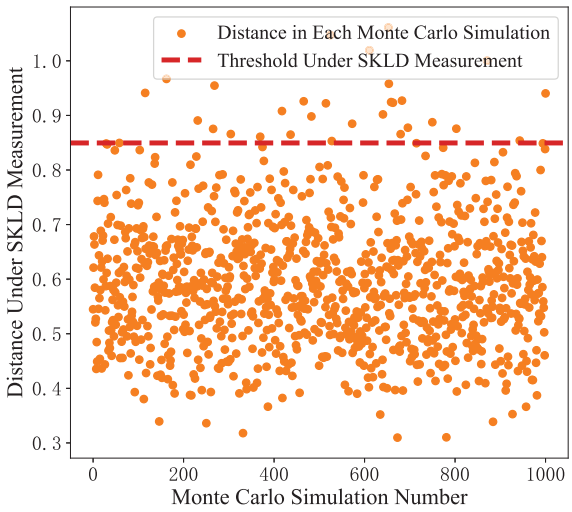
C. JR-DTC and Correct Judgment Probability Under AIRM and SKLD Metrics

To evaluate the performance of the proposed interference analysis method based on manifold DTC in the presence of potential interference from multiple "Starlink" satellites, four possible interference scenarios involving two "Starlink" satellites are considered as examples. These scenarios include: no interference from either "Starlink" satellite, interference only from "Starlink" satellite 1, interference only from "Starlink" satellite 2, and interference from both "Starlink" satellites. These four interference scenarios are labeled as Case 0, Case

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.



(a) The number of sampling points is 15.



(b) The number of sampling points is 100.

Fig. 5. The DTCs and threshold values obtained under the SKLD metric.

1, Case 2, and Case 3, respectively. The performance of the method under these conditions is analyzed and compared with energy-based method.

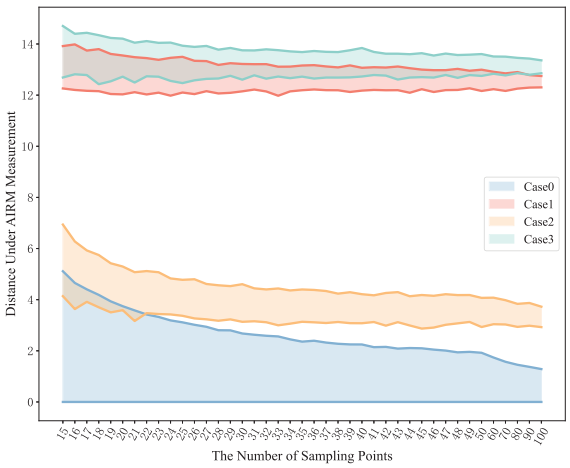
By substituting the parameters related to "OneWeb" and "Starlink" satellites from TABLE. I into (14) and (15), the power of the signals received by "OneWeb" Earth station under interference-free and three interference scenarios are obtained. The amplitude of the received signal is then calculated and substituted into (1) and (2). For each number of sampling points, 10 independent samplings were performed, and the data from these 10 independent samplings are substituted into (18) and (22) to obtain 10×10 covariance matrices. This allows the construction of points on the manifolds defined by (19) and (23). Subsequently, using the "center" matrices of the manifold obtained under the AIRM and SKLD metrics from Section IV. A and IV.B, the DTCs of the covariance matrices under interference-free and three interference scenarios are obtained using (26) and (27). Finally, (29) is employed to determine the presence or absence of interference and to distinguish among different interference scenarios.

The 1000 DTCs obtained under various interference scenarios are sorted in ascending order. With a false alarm rate set at 0.03, the values of QP_f and $Q(1-P_f)$ are determined to be 30 and 970, respectively. This means that the upper and lower bounds of the DTCs for each interference scenario correspond to the 970th and 30th DTCs in the sorted list. Based on this, the JR-DTC for each interference scenario is derived. Using the number of sampling points as the horizontal axis and the DTC as the vertical axis, different colors are filled between the upper and lower bounds of the DTCs to represent the JR-DTCs under different interference scenarios, as illustrated in Fig. 6. In Fig. 6, the areas covered by blue, red, orange, and green represent the JR-DTC correspond to Case0, Case1, Case2, Case3, respectively. In the Figure, the curves are colored in the order of blue, red, orange, and green, with each color representing the upper and lower bounds of the corresponding JR-DTC.

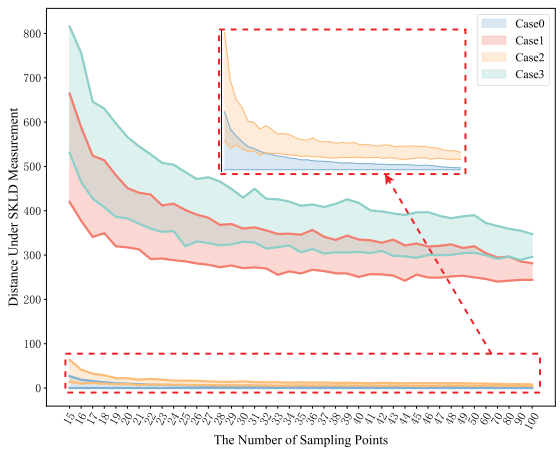
To verify the performance of the method proposed in this paper, after obtaining the JR-DTC, combining the signal model in four interference cases, 10000 Monte Carlo simulations are carried out to calculate the interference detection probabilities of the proposed method under AIRM and SKLD metrics and the energy-based method for different interference scenarios. In each simulation, points are randomly sampled from the signal models of four interference cases, and the sampling times of each interference cases are recorded. Then, the covariance matrix of the sampled data is brought into (24) and (25) respectively to calculate the DTC. If the DTC belongs only to the JR-DTC corresponding to the interference scenario of the sampled signal model and not to the JR-DTCs corresponding to the other three interference scenarios, then the interference scenario is considered correctly identified, and the number of correct identifications for the four interference scenarios is recorded. Finally, the correct judgment probability for each interference scenario is obtained by dividing the number of correct judgments by the total number of trials. With sampling points as the horizontal axis and correct judgment probability as the vertical axis, the variation curve of correct judgment probability with sampling points is drawn, as shown in Fig. 7.

In Fig. 7, the pentagram, triangle, and solid circle represent the curves of detection probability versus the number of sampling points under the AIRM metric, the SKLD metric, and the energy-based method across various interference scenarios, respectively. According to the aforementioned definition of interference detection probability, the three curves corresponding to Case0 in Fig. 7 represent the interference detection probability and correct judgment probability under the two metrics and the energy-based method in the absence of interference. As can be seen from the figure, when the number of sampling points is below 18, the proposed method achieves the highest detection probability for Case0 among the four interference scenarios, but it is significantly lower than that of the energy-based method. As the number of sampling points increases, the detection probability of the proposed method for all four scenarios increases significantly, while the detection probability of the energy-based method increases more slowly. When the number of sampling points exceeds 18, 21, 50, and 50, respectively, the detection probability of the proposed

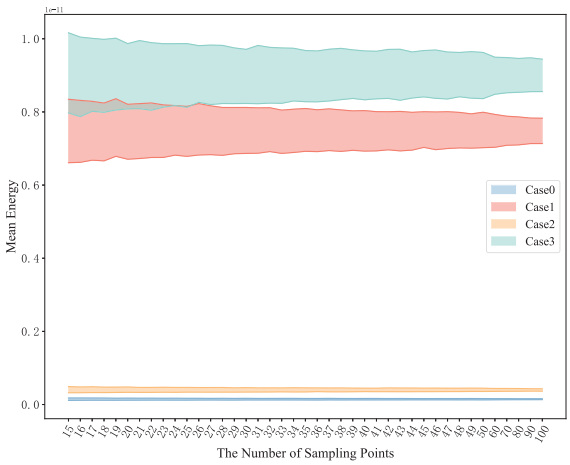
Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.



(a) Under AIRM metric.



(b) Under SKLD metric.



(c) Energy-based method.

Fig. 6. The variation trend of JR-DTC when the number of sampling points is 15-50,60,70,80,90,100.

method for Case2, Case0, Case3, and Case1 surpasses that of the energy-based method. When the number of sampling points exceeds 80, the detection probabilities for all scenarios tend to stabilize, with the detection probability for Case0 exceeding 95% and those for Case1, Case2, and Case3 exceeding 99%. Based on the above analysis, it can be concluded that when the number of sampling points exceeds 50, the detection performance of the proposed method is superior to that of the energy-based method in all scenarios, enabling highly accurate detection of the presence or absence of interference and discriminative analysis of different interference scenarios.

D. Visualization of Interference Cases

Assume that the longitude and latitude coordinates of a subsatellite point of a "Starlink" satellite vary from 110°E to 120°E and 35°N to 45°N respectively, and assume that three "OneWeb" system's Earth stations are deployed within the longitude and latitude range, located at (40.12°N,116.23°E), (40°N,112°E), and (43°N,114°E), respectively. All three "OneWeb" Earth stations point to the "OneWeb" satellite. In order to keep the difference between adjacent data from being too large and avoid the error caused by the large uncertainty of the sampling data, the precision of the latitude and longitude grid is set to 0.01°, the independent sampling times are still 10 times, and the sampling points are changed to 500.

When the "Starlink" satellite is located on different latitude and longitude grids, the interference power of the "Starlink" satellite to the three "OneWeb" Earth stations are calculated according to (15), and the amplitude is obtained. Then, the amplitude is brought into (2) and a series of points obeying the model of interference signal in (2) are generated by the computer. By sampling these points 10 times independently with 500 sampling points each time, a 1000×1000 covariance matrix can be obtained for each Earth station. By bringing these covariance matrices and the number of independent samples into (28), the potential function of the sum of the interference caused by "Starlink" satellites on the three "OneWeb" Earth stations at different latitude and longitude grids is obtained. It can be seen from (20) that when the communication parameters of "Starlink" satellite are determined, the value of the covariance matrix of the sampled data is only related to the longitude and latitude of the subsatellite points of "Starlink" satellite. Therefore, the longitude and latitude of the subsatellite points of "Starlink" can be taken as horizontal and vertical coordinates, and a three-dimensional coordinate system can be established with the potential function value of the sum of interference caused by "Starlink" satellite to the three "OneWeb" Earth stations as z-axis coordinates, and a three-dimensional surface diagram of "latitude-longitude-potential function" can be obtained, as shown in Fig. 8.

As can be seen from Fig. 8, the potential function of the sum of the sampled data of the received signals of the three "OneWeb" Earth stations peaks around the longitude and latitude of the three "OneWeb" Earth stations, namely (40.12°N,116.23°E), (38°N,112°E), and (42°N,114°E). At the same time, a peak also occurred near the latitude and longitude of the "OneWeb" satellite, i.e. (35.773°N,113.965°E), and the peak near this point was higher than the peak of the previous three points. This is because the closer the "Starlink" satellite gets to the "OneWeb" Earth station, the less the link loss between it and the "OneWeb" Earth station, and therefore the higher power it receives from the signal. Since the three "OneWeb" Earth stations all point to the "OneWeb" satellite,

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

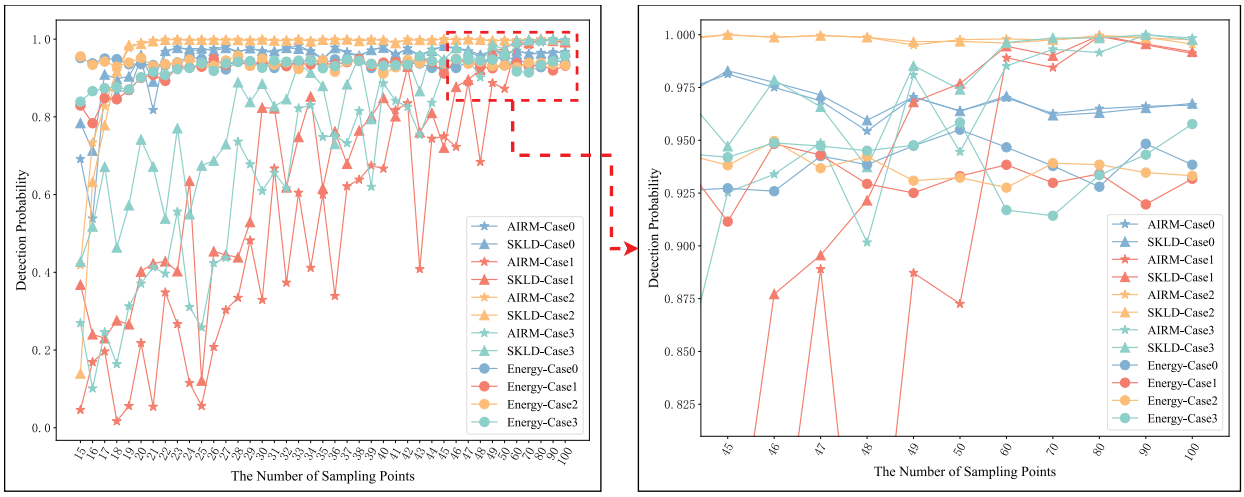


Fig. 7. The variation curve of correct judgment probability of four interference cases with sampling points.

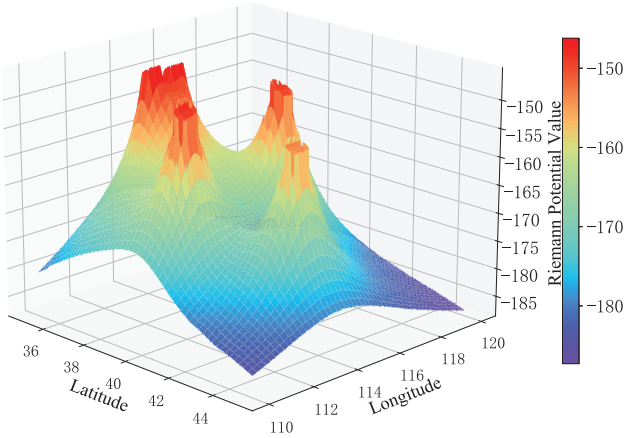


Fig. 8. Affine embedding representation of the sum of interference from a single "Starlink" satellite to three "OneWeb" Earth stations in three-dimensional space.

when the "Starlink" satellite is closer to the "OneWeb" satellite, the reception gain of the "OneWeb" Earth station will be larger, so the three-dimensional surface diagram of "latitude-longitude-potential function" also appears a peak near the "OneWeb" satellite.

V. Conclusion

This paper addressed the problem of real-time and accurate analysis of CFI between NGSO constellation systems. Based on information geometry theory, we considered the probability distribution characteristics of the received sampled signals as the starting point and shift the traditional Euclidean space-based interference analysis to matrix manifolds. We established the manifold geometric structure and parametric characterization model of the probability distribution of downlink CFI between NGSO constellation systems and proposed an interference analysis method based on the DTC of manifolds. The detection performance of this method was analyzed through simulations under both the AIRM and SKLD metrics, and it is compared with the energy-based method. Finally, the signals received by the "OneWeb" Earth station were visually represented in three-dimensional space by affine embedding method. The results showed that once the interference analysis model of this method is established, the JR-DTC can be utilized to rapidly analyze interference scenarios. It achieved superior interference detection performance with few sampling points compared to the traditional energy-based method while effectively avoiding the complexity of interference analysis caused by the time-varying characteristics of NGSO constellation. This enables real-time and accurate judgment and analysis of CFI in mega NGSO constellations, providing a new and efficient approach for analyzing CFI between NGSO constellations and designing NGSO constellation system links. In future research, we need to further explore the impact of complex satellite communication channel models on the performance of the method, as well as its application in rapid CFI analysis for direct-to-smartphone communication [39].

References

[1] T. Butash, P. Garland, and B. Evans, "Non-geostationary satellite orbit communications satellite constellations history," *Int. J. Satell. Commun. Netw.*, vol. 39, no. 1, pp. 1–5, Aug. 2020.

[2] H. Al-Hraishawi, H. Chougrani, S. Kisseleff, E. Lagunas, and S. Chatzinotas, "A survey on nongeostationary satellite systems: The communication perspective," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 101–132, 1st Quart., 2023.

[3] P. Zhang, Z. Xian, M. Liao, H. Huang, and J. Yang, "Adaptive routing mechanism for LEO satellite network based on control domain partition," *IEEE Trans. Green Commun. Netw.*, vol. 9, no. 1, pp. 70–82, Mar. 2025.

[4] C.-S. Choi, "Modeling and analysis of downlink communications in a heterogeneous LEO satellite network," *IEEE Trans. Wireless Commun.*, vol. 23, no. 8, pp. 8588–8602, Aug. 2024.

[5] Y. He, Y. Liu, C. Jiang, and X. Zhong, "Multiobjective anti-collision for massive access ranging in MF-TDMA satellite communication system," *IEEE Internet Things J.*, vol. 9, no. 16, pp. 14655–14666, Aug. 2022.

[6] J. Liu, Y. Shi, Z. M. Fadlullah, and N. Kato, "Space-air-ground integrated network: A survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2714–2741, 4th Quart., 2018.

[7] Y. He, B. Sheng, H. Yin, Y. Liu, and Y. Zhang, "Distributed satellite cluster laser networking algorithm with double-layer Markov DRL architecture," *Space, Sci. Technol.*, vol. 3, Jan. 2023, Art. no. 0012.

[8] Z. M. Bakhsh, Y. Omid, G. Chen, F. Kayhan, Y. Ma, and R. Tafazolli, "Multi-satellite MIMO systems for direct satellite-to-device communications: A Survey," *IEEE Commun. Surveys Tuts.*, early access, Aug. 26, 2024, doi: 10.1109/COMST.2024.3449430.

[9] L. Wei, W. Wenkang, L. Chang, L. Shanshan, and W. Xiuguang, "Interference analysis simulation method based on spatial position probability for NGSO communication constellation system," *Chin. J. Radio Sci.*, vol. 36, no. 3, pp. 483–490, 2021.

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.

[10] C. Li, L. Zhu, C. Guo, T. Liu, and Z. Zhang, "Intelligent blind source separation technology based on OTFS modulation for LEO satellite communication," *China Commun.*, vol. 19, no. 7, pp. 89–99, Jul. 2022.
[11] Y. He, Y. Li, and H. Yin, "Co-frequency interference analysis and avoidance between NGSO constellations: Challenges, techniques, and trends," *China Commun.*, vol. 20, no. 7, pp. 1–14, Jul. 2023.
[12] E. Kim, I. P. Roberts, and J. G. Andrews, "Feasibility analysis of in-band coexistence in dense LEO satellite communication systems," *IEEE Trans. Wireless Commun.*, vol. 24, no. 2, pp. 1663–1677, Feb. 2025.
[13] C.-S. Park, C.-G. Kang, Y.-S. Choi, and C.-H. Oh, "Interference analysis of geostationary satellite networks in the presence of moving non-geostationary satellites," in *Proc. 2nd Int. Conf. Inf. Technol. Converg. Services*, Aug. 2010, pp. 1–5.
[14] H. Wang, C. Wang, J. Yuan, Y. Zhao, R. Ding, and W. Wang, "Coexistence downlink interference analysis between LEO system and GEO system in Ka band," in *Proc. IEEE Int. Conf. Commun.*, Beijing, China, Aug. 2018, pp. 465–469.
[15] C. Dai, P. Li, H. Fang, and H. Guo, "Interference and coordination analysis of non-geostationary orbit satellites," *Modern Electron. Technique*, vol. 44, no. 21, pp. 1–5, Sep. 2021.
[16] J. Jin, L. Yaqiang, Z. Chen, K. Linling, and Y. Jian, "Occurrence probability of co-frequency interference and system availability of non-geostationary satellite system in global dynamic scene," *J. Tsinghua Univ.*, vol. 58, no. 9, pp. 833–840, 2018.
[17] D. Yan, Y. He, and H. Fu, "Interference analysis of NGSO constellation to GEO satellite communication system based on spatio-temporal slices," *IEEE Internet Things J.*, vol. 10, no. 18, pp. 16605–16616, Sep. 2023.
[18] H. Jia, Z. Ni, C. Jiang, L. Kuang, and J. Lu, "Uplink interference and performance analysis for megasatellite constellation," *IEEE Internet Things J.*, vol. 9, no. 6, pp. 4318–4329, Mar. 2022.
[19] Z. Hongshi and B. Jiang, "Spatial isolation methodology for spectral coexistence in LEO satellite systems," *J. Beijing Univ. Aeronaut. Astronaut.*, vol. 44, no. 9, p. 1909, Sep. 2018.
[20] F. Peng et al., "Study on system-level interference evaluation of non-geostationary satellite constellation," *Chin. Space Sci. Technol.*, vol. 43, no. 5, pp. 46–55, Sep. 2023.
[21] W. Yu, Y. Cheng, and W. Li, "Simulation research of frequency interference for low Earth orbit satellite constellation networks," *J. CAEIT*, vol. 5, pp. 501–507, Sep. 2022.
[22] S. Amari and H. Nagaoka, "Methods of information geometry," *Amer. Math. Soc.*, vol. 191, no. 13, pp. 115–127, Apr. 2000.
[23] F. Nielsen and F. Barbaresco, *Geometric Science of Information*, vol. 14071. Cham, Switzerland: Springer.
[24] H. Wang, X. Li, Y. Cheng, and Y. Qin, "Progress in theory and applications of information geometry," *Scientia Sinica Informationis*, vol. 43, no. 6, pp. 707–732, Jun. 2013.
[25] H. Sun, L. Peng, and Z. Zhang, "Information geometry and its applications," *Adv. Math.*, vol. 191, no. 4, pp. 257–269, Oct. 2011.
[26] A. Morello and V. Mignone, "DVB-s2: The second generation standard for satellite broad-band services," *Proc. IEEE*, vol. 94, no. 1, pp. 210–227, Jan. 2006.
[27] *Satellite Antenna Radiation Patterns for Non-geostationary Orbit Satellite Antennas Operating in the Fixed-satellite Service Below 30 GHz*, document ITU-R S.1528, 2001.
[28] *Reference Radiation Pattern of Earth Station Antennas in the Fixedsatellite Service for Use in Coordination and Interference Assessment in the Frequency Range from 2 to 31 GHz*, document ITU-R S.465, 2010.
[29] *Specific Attenuation Model for Rain for Use in Prediction Methods*, document ITU-R P.838-3, 2005.
[30] *Attenuation Due To Clouds Fog*, document ITU-R P.840-7, 2017.
[31] N. Ay, P. Gibilisco, and F. Matús, "Information geometry and its applications," in *Springer Proceedings in Mathematics & Statistics*, vol. 252. Cham, Switzerland: Springer, Jun. 2016.
[32] Q. Chen, P. Wan, Y. Wang, J. Li, and Y. Xiao, "Research on cognitive radio spectrum sensing method based on information geometry," in *Proc. Int. Conf. Cloud Comput. Secur.*, vol. 10603, Jan. 2017, pp. 554–564.
[33] X. Hua, Y. Cheng, H. Wang, Y. Wang, and L. Zhang, "Matrix information geometric detectors with manifold filter," *J. Nat. Univ. Defense Technol.*, vol. 44, no. 6, pp. 51–60, 2022.
[34] J. Yang, Y. Cheng, H. Wang, Y. Li, and X. Hua, "Unknown stochastic signal detection via non-Gaussian noise modeling," in *Proc. IEEE Int. Conf. Signal Process., Commun. Comput. (ICSPCC)*, Ningbo, China, Sep. 2015, pp. 1–4.
[35] Y. Cheng, X. Wang, M. Morelande, and B. Moran, "Information geometry of target tracking sensor networks," *Inf. Fusion*, vol. 14, no. 3, pp. 311–326, Jul. 2013.
[36] C. Zhang, C. Jiang, J. Jin, S. Wu, L. Kuang, and S. Guo, "Spatial angular spectrum sensing for non-geostationary satellite systems," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Abu Dhabi, UAE, Jul. 2018, pp. 1–6.
[37] K. Arwini and C. Dodson, *Information Geometry: Near Randomness and Near Independence*, vol. 1953. Cham, Switzerland: Springer, 2008.
[38] K.-X. Chen, J.-Y. Ren, X.-J. Wu, and J. Kittler, "Covariance descriptors on a Gaussian manifold and their application to image set classification," *Pattern Recognit.*, vol. 107, Nov. 2020, Art. no. 107463.
[39] Y. He, Y. Xiao, S. Zhang, M. Jia, and Z. Li, "Direct-to-smartphone for 6G NTN: Technical routes, challenges, and key technologies," *IEEE Netw.*, vol. 38, no. 4, pp. 128–135, Jul. 2024.

Yuanzhi He received the Ph.D. degree from the PLA University of Science and Technology, Nanjing, China. She is currently a Researcher with PLA, Beijing, China, and an Adjunct Professor with Beihang University, Beijing Institute of Technology, and Sun Yat-sen University. Her current research interests include satellite communication systems, space information networks, and space-air-ground integrated networks.

Di Yan received the B.S. degree in communication and information systems from the PLA Army Aviation Institute, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree with the Institute of Systems Engineering, Military Academy of Science, Beijing. His current research interests include satellite communication systems and the satellite IoT.

Chengwu Qi received the B.S. degree in communication and information systems from the PLA Army Aviation Institute, Beijing, China, in 2022. He is currently pursuing the M.S. degree with the Institute of Systems Engineering, Academy of Military Sciences, Beijing. His current research interests include satellite communication system and interference analysis.

Authorized licensed use limited to: BEIJING UNIVERSITY OF POST AND TELECOM. Downloaded on November 21,2025 at 01:42:55 UTC from IEEE Xplore. Restrictions apply.